This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-01** Sheet 1 of 1

OPL Title	**Lube Oil System Pre-Start & Priming**	**Discipline**	Mechanical
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **lube oil system pre-start & priming** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. With no lube oil pressure the crankshaft bearings fail in seconds; the pre-lube pump must build pressure before the machine is cranked. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Run the auxiliary lube oil pump and build pressure above 2 barg	Isolation & LOTO verified
2	Confirm PSLL-4504 shows healthy oil pressure	Parts match BOM / spec
3	Check the frame oil level and the filter differential pressure	Torque / clearance to spec
4	Verify no oil leaks and correct oil grade in the frame	No leak / smooth operation
5	Only then give the start command to the main motor	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- With no lube oil pressure the crankshaft bearings fail in seconds; the pre-lube pump must build pressure before the machine is cranked.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Friday, 24 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-02** Sheet 1 of 1

OPL Title	**Compressor Suction/Discharge Valve Inspection**	**Discipline**	Mechanical
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **compressor suction/discharge valve inspection** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Broken plate valves are the classic cause of high discharge temperature (TSHH-4503) and lost capacity on KC-4501. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate, depressurise and purge the cylinder with nitrogen	Isolation & LOTO verified
2	Remove the suction and discharge valve caps	Parts match BOM / spec
3	Inspect valve plates, springs and seats; replace as a complete set	Torque / clearance to spec
4	Check the valve gasket and seat for damage	No leak / smooth operation
5	Torque the valve caps to spec and leak test before start	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Broken plate valves are the classic cause of high discharge temperature (TSHH-4503) and lost capacity on KC-4501.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Tuesday, 28 April 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-03** Sheet 1 of 1

OPL Title	**Seal Gas (N2 Buffer) Management**	**Discipline**	Process / Operations
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **seal gas (n2 buffer) management** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. The nitrogen buffer keeps flammable process gas out of the crankcase; loss of seal gas is a fire/explosion risk. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Verify FT-4506 seal gas flow is established before start	Permit valid & briefed
2	Maintain the buffer differential above process pressure	Parameter within limit
3	Watch the distance-piece vent for signs of leakage	Interlock status confirmed
4	Respond immediately to a seal-gas low alarm	Change logged on DCS
5	Never run the compressor with seal gas lost	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- The nitrogen buffer keeps flammable process gas out of the crankcase; loss of seal gas is a fire/explosion risk.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Saturday, 02 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-04** Sheet 1 of 1

OPL Title	**Anti-Surge / Recycle Operation**	**Discipline**	Process / Operations
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **anti-surge / recycle operation** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. The recycle (spillback) valve protects the machine at low throughput and during start/stop; incorrect operation risks deadhead and rod overload. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Keep the anti-surge FV-4502 in auto during start-up	Permit valid & briefed
2	Avoid deadheading against a high discharge pressure (PSHH-4502)	Parameter within limit
3	Watch suction PSLL-4501 for gas starvation	Interlock status confirmed
4	Return the machine to load slowly, closing recycle gradually	Change logged on DCS
5	Investigate any repeated surge cycling with the vendor curve	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- The recycle (spillback) valve protects the machine at low throughput and during start/stop; incorrect operation risks deadhead and rod overload.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Wednesday, 06 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-05** Sheet 1 of 1

OPL Title	**Crosshead Vibration Monitoring (VSHH-4505)**	**Discipline**	Instrument
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **crosshead vibration monitoring (vshh-4505)** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. A rising crosshead vibration trend warns of loose running gear before it trips or fails. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trend VT-4501 crosshead vibration on the daily route	Loop isolated / under permit
2	Alarm at 8 mm/s and trip at 11 mm/s (VSHH-4505)	Reference standard traceable
3	Correlate any rise with TE-4507 main bearing temperature	As-found & as-left recorded
4	Investigate rod-load reversals and crosshead pin clearance	Trip / control action verified
5	Escalate a sudden step-change to reliability immediately	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Proof test of lube oil low-low trip	Set-point within tolerance	Recalibrated PSLL-4504 to 1.5 barg, function tested
Seal gas flow FT-4506 low, buffer pressure unstable	N2 buffer regulator diaphragm failed	Replaced seal gas regulator, re-established buffer dP
Proof test of KO drum high-high level trip	Float healthy, minor set-point trim	Verified LSHH-4510 float and trip action

6. KEY LEARNING POINTS

- A rising crosshead vibration trend warns of loose running gear before it trips or fails.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Sunday, 10 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-06** Sheet 1 of 1

OPL Title	**Suction KO Drum Draining (FA-4510)**	**Discipline**	Process / Operations
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **suction ko drum draining (fa-4510)** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Liquid carried over from the KO drum destroys valves and can bend piston rods; the boot must be drained regularly. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Drain the FA-4510 boot every shift on the operator round	Permit valid & briefed
2	Watch LT-4510 level and the LSHH-4510 high-high trip	Parameter within limit
3	Never run the compressor with a high KO drum level	Interlock status confirmed
4	Report frequent liquid carryover to process for investigation	Change logged on DCS
5	Confirm the auto-drain (if fitted) is functioning	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Liquid carried over from the KO drum destroys valves and can bend piston rods; the boot must be drained regularly.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Thursday, 14 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-KC-4501-07** Sheet 1 of 1

OPL Title	**Intercooler EA-4502 Performance Check**	**Discipline**	Mechanical
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	**Area / Unit**	4500 - RECYCLE GAS SYSTEM
Related Interlock	SEQ-4501 (SIL 2)	**P&ID Ref**	TJC-LLD-PID-4501

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **intercooler ea-4502 performance check** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Poor intercooling raises the 2nd-stage suction temperature and hence the final discharge temperature towards the trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the interstage temperature drop across EA-4502	Isolation & LOTO verified
2	Verify the jacket water flow FSL-4508 is adequate	Parts match BOM / spec
3	Clean the intercooler tubes if the approach temperature rises	Torque / clearance to spec
4	Log the interstage pressure and compare to datasheet	No leak / smooth operation
5	Report fouling trends so cleaning can be planned	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Poor intercooling raises the 2nd-stage suction temperature and hence the final discharge temperature towards the trip.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Monday, 18 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-01** Sheet 1 of 1

OPL Title	**Tube-Side Cleaning (Hydrojetting)**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **tube-side cleaning (hydrojetting)** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Hexane fouling on the tube side raises the differential pressure (PDT-5605) and cuts the heat duty, leaving cold solvent downstream. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and drain both sides, then purge with nitrogen	Isolation & LOTO verified
2	Open the channel cover and remove the old gasket	Parts match BOM / spec
3	Hydrojet the tube bores and inspect for pitting/erosion	Torque / clearance to spec
4	Verify tube-side dP restored and no residual blockage	No leak / smooth operation
5	Re-gasket with new SW SS316 gasket and box up to torque	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Hexane fouling on the tube side raises the differential pressure (PDT-5605) and cuts the heat duty, leaving cold solvent downstream.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Tuesday, 05 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-02** Sheet 1 of 1

OPL Title	**Gasket Replacement & Flange Box-Up**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gasket replacement & flange box-up** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Spiral-wound gaskets are single-use; reusing one causes hexane leaks at the channel and shell flanges. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Remove the old shell and channel gaskets completely	Isolation & LOTO verified
2	Clean and inspect the gasket seating faces for damage	Parts match BOM / spec
3	Fit new SW SS316/graphite gaskets, correctly centred	Torque / clearance to spec
4	Torque the flange bolts in a cross pattern in passes	No leak / smooth operation
5	Leak test on start-up and re-torque hot if permitted	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Spiral-wound gaskets are single-use; reusing one causes hexane leaks at the channel and shell flanges.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Saturday, 09 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-03** Sheet 1 of 1

OPL Title	**Steam Trap Health Check**	**Discipline**	Process / Operations
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **steam trap health check** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. A failed steam trap either floods the shell (killing duty) or blows live steam (wasting energy); regular checks catch both. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Compare the condensate temperature TI-5604 to steam temperature	Permit valid & briefed
2	Use ultrasonic or a listening stick to check trap operation	Parameter within limit
3	A cold trap means flooding; a continuously hot blow means leak-through	Interlock status confirmed
4	Check the condensate pot level gauge LG-5606	Change logged on DCS
5	Replace the trap element or trap as required	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- A failed steam trap either floods the shell (killing duty) or blows live steam (wasting energy); regular checks catch both.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Wednesday, 13 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-04** Sheet 1 of 1

OPL Title	**Thermal-Shock Avoidance on Start-Up**	**Discipline**	Process / Operations
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **thermal-shock avoidance on start-up** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Admitting steam onto a cold tube bundle too fast cracks the tube-to-tubesheet joints, causing internal leaks. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Establish solvent (tube-side) flow first	Permit valid & briefed
2	Admit steam slowly by ramping the TV-5602 / TIC-5602 set-point	Parameter within limit
3	Warm the exchanger through gradually	Interlock status confirmed
4	Avoid sudden large steps in the outlet temperature set-point	Change logged on DCS
5	Watch for knocking that indicates condensate hammer	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- Admitting steam onto a cold tube bundle too fast cracks the tube-to-tubesheet joints, causing internal leaks.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Sunday, 17 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-05** Sheet 1 of 1

OPL Title	**Tube Leak Detection & Plugging**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **tube leak detection & plugging** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. A tube leak lets steam into the solvent (rising outlet temperature and water in the solvent); the leaking tube must be found and plugged. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and hydrotest the shell side to find the leak	Isolation & LOTO verified
2	Bubble-test to identify the specific leaking tube	Parts match BOM / spec
3	Plug both ends of the tube with tapered plugs	Torque / clearance to spec
4	Record the plugged-tube count (maximum 5 % of tubes)	No leak / smooth operation
5	Plan bundle replacement if the plugging limit is reached	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- A tube leak lets steam into the solvent (rising outlet temperature and water in the solvent); the leaking tube must be found and plugged.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Thursday, 21 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-06** Sheet 1 of 1

OPL Title	**Fouling Trend & Cleaning Schedule (PDT-5605)**	**Discipline**	Instrument
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fouling trend & cleaning schedule (pdt-5605)** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. Trending the tube-side differential pressure predicts when cleaning is due before the duty is lost. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Log PDT-5605 differential pressure weekly	Loop isolated / under permit
2	Note the PDAH-5605 fouling alarm at 0.7 bar dP	Reference standard traceable
3	Correlate rising dP with any outlet-temperature shortfall	As-found & as-left recorded
4	Plan cleaning before the duty loss becomes critical	Trip / control action verified
5	Feed the trend into the exchanger cleaning program	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Outlet temperature hunting around set-point	Steam valve TV-5602 loop poorly tuned	Retuned TIC-5602 PID, verified stable control
Erratic tube dP reading	Impulse lines plugged with fines	Cleaned and re-zeroed PDT-5605 transmitter
Cracked gauge glass on condensate pot	Thermal/mechanical stress on glass	Replaced LG-5606 gauge glass and seals

6. KEY LEARNING POINTS

- Trending the tube-side differential pressure predicts when cleaning is due before the duty is lost.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Monday, 25 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-EA-5601-07** Sheet 1 of 1

OPL Title	**PSV-5607 Testing & Car-Seal Management**	**Discipline**	Mechanical
Equipment	EA-5601 - SOLVENT HEATER	**Area / Unit**	5600 - SOLVENT HEATING SYSTEM
Related Interlock	N/A (control loop only) (N/A)	**P&ID Ref**	TJC-LLD-PID-5601

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **psv-5607 testing & car-seal management** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. The tube-side relief valve is the exchanger's overpressure protection and must be proven and never isolated in service. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Confirm PSV-5607 is set to 16 barg on the nameplate	Isolation & LOTO verified
2	Carry out an off-line pop test per the statutory schedule	Parts match BOM / spec
3	Keep the PSV inlet/outlet isolation car-sealed open in service	Torque / clearance to spec
4	Record the test certificate and next due date	No leak / smooth operation
5	Never block-in a live PSV without an authorised spare	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- The tube-side relief valve is the exchanger's overpressure protection and must be proven and never isolated in service.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Arya Wibisono (EMP-0912)	Friday, 29 May 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-01** Sheet 1 of 1

OPL Title	**Control Valve Stroke Calibration (DVC6200)**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **control valve stroke calibration (dvc6200)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Correct mapping of the 4-20 mA signal to 0-100 % valve stroke is essential for stable separator level control. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Connect ValveLink to the DVC6200 positioner	Loop isolated / under permit
2	Run the auto-calibration over the full stroke	Reference standard traceable
3	Verify 4 mA = closed and 20 mA = 38 mm fully open	As-found & as-left recorded
4	Check hysteresis and dead-band are within 1 %	Trip / control action verified
5	Save the calibration and record the as-left data	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Correct mapping of the 4-20 mA signal to 0-100 % valve stroke is essential for stable separator level control.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Saturday, 16 May 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-02** Sheet 1 of 1

OPL Title	**Instrument Air Supply & Filter Regulator**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **instrument air supply & filter regulator** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Dirty or low instrument air makes the valve sluggish or fail closed, upsetting FA-6710 level and tripping PSL-6702. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the PI-6702 air supply is at 1.4 barg	Loop isolated / under permit
2	Drain the filter regulator bowl of condensate	Reference standard traceable
3	Replace a clogged air filter element	As-found & as-left recorded
4	Confirm there are no air leaks on the tubing/fittings	Trip / control action verified
5	Verify PSL-6702 low-air alarm set-point	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Dirty or low instrument air makes the valve sluggish or fail closed, upsetting FA-6710 level and tripping PSL-6702.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Wednesday, 20 May 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-03** Sheet 1 of 1

OPL Title	**Packing Adjustment (PTFE V-Ring)**	**Discipline**	Mechanical
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[ ] Improvement	[X] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **packing adjustment (ptfe v-ring)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Loose packing leaks hexane at the stem; over-tight packing makes the stem stick and the valve hunt. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate the line and depressurise before touching packing	Isolation & LOTO verified
2	Snug the packing nuts evenly in small increments	Parts match BOM / spec
3	Stroke the valve and check for a leak-free, smooth travel	Torque / clearance to spec
4	Do not over-torque - excessive friction degrades control	No leak / smooth operation
5	Record the packing adjustment in the valve history	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Manual handwheel jack on LV-6701 seized	Corrosion in handwheel thread	Overhauled handwheel assembly, greased thread
LV-6701 passing when closed (Class IV exceeded)	Seat/plug erosion from polymer bottoms	Lapped seat and plug, restored Class IV shut-off
Minor weep at body/bonnet joint	Spiral-wound body gasket relaxed	Renewed body gasket, torqued bonnet bolts

6. KEY LEARNING POINTS

- Loose packing leaks hexane at the stem; over-tight packing makes the stem stick and the valve hunt.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Sunday, 24 May 2026

P&ID

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-04** Sheet 1 of 1

OPL Title	**Positioner Response & Loop Check**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **positioner response & loop check** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Poor positioner tuning makes the level controller LIC-6701 hunt and the separator level cycle. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Step the LIC-6701 output and watch the valve response	Loop isolated / under permit
2	Check the ZSL/ZSH-6701 limit-switch feedback	Reference standard traceable
3	Tune the positioner gain for a stable, prompt travel	As-found & as-left recorded
4	Confirm there is no overshoot or stiction step	Trip / control action verified
5	Verify the valve seats fully at 4 mA (Class IV)	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Poor positioner tuning makes the level controller LIC-6701 hunt and the separator level cycle.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Thursday, 28 May 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-05** Sheet 1 of 1

OPL Title	**Manual Bypass (HV-6701) Operation**	**Discipline**	Process / Operations
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manual bypass (hv-6701) operation** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. The manual bypass lets operators keep the separator level safely while LV-6701 is isolated for maintenance. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Notify the panel and take LIC-6701 to manual	Permit valid & briefed
2	Crack open the bypass HV-6701 slowly	Parameter within limit
3	Throttle HV-6701 to hold FA-6710 level using LT-6710	Interlock status confirmed
4	Isolate LV-6701 (close block valves) for the work	Change logged on DCS
5	Reverse the steps and re-establish auto control afterwards	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 isolation for planned maintenance	Planned - bypass used to hold level	Lined up HV-6701 manual bypass, isolated LV-6701 safely
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Manual handwheel jack on LV-6701 seized	Corrosion in handwheel thread	Overhauled handwheel assembly, greased thread

6. KEY LEARNING POINTS

- The manual bypass lets operators keep the separator level safely while LV-6701 is isolated for maintenance.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Monday, 01 June 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-06** Sheet 1 of 1

OPL Title	**Fail-Closed & ESD Action Verification**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[X] Basic Knowledge	[ ] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fail-closed & esd action verification** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. On air failure or ESD (XV-6701), the valve must drive closed to prevent gas blow-by from the separator. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trip the XV-6701 ESD solenoid during the test	Loop isolated / under permit
2	Confirm the actuator vents and the valve closes	Reference standard traceable
3	Time the full closure - it must be within 5 seconds	As-found & as-left recorded
4	Check the ZSL-6701 closed limit confirms on the DCS	Trip / control action verified
5	Reset the ESD and return the valve to auto	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- On air failure or ESD (XV-6701), the valve must drive closed to prevent gas blow-by from the separator.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Friday, 05 June 2026

This is sample data provided for CALIBER purposes only	**ONE POINT LESSON (OPL)**	OPL No: **OPL-LV-6701-07** Sheet 1 of 1

OPL Title	**Actuator Diaphragm Inspection (Fisher 667)**	**Discipline**	Instrument
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	**Area / Unit**	6700 - LP SEPARATOR SYSTEM
Related Interlock	SEQ-6701 (SIL 1)	P&ID **Ref**	TJC-LLD-PID-6701

Classification:	[ ] Basic Knowledge	[X] Improvement	[ ] Trouble Case	Aspect: P-Q-C-D-**S**-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **actuator diaphragm inspection (fisher 667)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. A ruptured actuator diaphragm causes loss of control and a continuous air leak; periodic inspection prevents surprise failures. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inspect the Fisher 667 diaphragm for cracks or perishing	Loop isolated / under permit
2	Check the bench set is 0.2-1.0 barg	Reference standard traceable
3	Replace the NBR/fabric diaphragm if hardened	As-found & as-left recorded
4	Re-stroke and recalibrate after any diaphragm work	Trip / control action verified
5	Confirm the spring range still gives full 38 mm travel	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- A ruptured actuator diaphragm causes loss of control and a continuous air leak; periodic inspection prevents surprise failures.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Vino Ardiansyah (EMP-1102)	Zulkarnain Hasan (EMP-0901)	Tuesday, 09 June 2026

Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.